

A. Arul Ananth

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SUMMARY

Third-year Computer Science undergraduate with hands-on experience in full-stack development, backend engineering, AI-powered applications, and AI-assisted software development. Strongly focused on learning and applying modern engineering practices across Agentic AI, RAG, local LLMs, orchestration, secure backend systems, and AI-led developer productivity.

Practically experienced in Java, Spring Boot, React, Python, FastAPI, MySQL, Qdrant, CrewAI, Ollama, and vector search. Actively uses AI coding assistants such as Codex / Claude-style agentic development tools for code generation, debugging, refactoring, documentation, test ideation, and productivity improvement, with human review and validation of generated code.

Passionate about solving emerging software challenges where AI can improve development speed, application intelligence, operational efficiency, and personalized user experiences.

EDUCATION

St. Joseph's College of Engineering, Chennai

B.E. Computer Science and Engineering, Current Academic Standing: Third-year undergraduate, CGPA: 8.52 2024 – 2028

D.A.V School, Adambakkam, Chennai

Class XII, 88.6%, Class X | 2022 | 92%

2024

EXPERIENCE

Developer Intern

Dec 2025

Sedin Technologies

- Contributed to a full-stack e-commerce platform by implementing role-based access control, product administration, and discount management features.
- Fixed authorization issues, security vulnerabilities, and application bugs to improve system stability, maintainability, and access control.
- Used AI-assisted development practices for understanding existing code, debugging issues, improving implementation speed, and validating changes before completion.
- Gained practical exposure to production-style software development, secure backend workflows, and real-world application maintenance.
- Tech Stack: React, Spring Boot, MySQL

Developer Intern

Jun 2025

Mindgate Solutions Pvt. Ltd.

- Developed a Data Report Application to improve report generation and structured data management processes.
- Built application features using Angular, Spring Boot, and MySQL.
- Applied backend API development, database interaction, and reporting workflow design in an enterprise application context.
- Strengthened understanding of how software systems support business reporting, operational visibility, and productivity.
- Tech Stack: Angular, Spring Boot, MySQL

PROJECTS

Lumeward – Hybrid AI Assistant

- Developed an AI-powered assistant for generating newsletters and research briefings using CrewAI agents and locally hosted LLMs through Ollama.
- Implemented document ingestion, embedding-based retrieval, and semantic search using Qdrant vector database.
- Built RAG-based retrieval flows to improve answer relevance using user-provided documents and personalized context.
- Designed a dual-runtime architecture that allows the application to run either as a standalone PySide6 desktop application or as a headless FastAPI server.
- Used agentic coding practices to accelerate implementation, explore design options, debug integration issues, and refine application workflows.
- Gained practical exposure to Agentic AI, orchestration, local LLM execution, document intelligence, and AI-led knowledge productivity.
- Tech Stack: Python, FastAPI, PySide6, CrewAI, Ollama, Mistral, Qdrant

ShopSphere – Full-Stack E-Commerce Platform

- Developed a full-stack e-commerce platform with product catalog management, shopping cart functionality, secure authentication, and role-based access control.
- Designed a modular monolith backend using Spring Boot, Spring Security, JWT-based stateless authentication, and Hibernate JPA.
- Built a responsive React and TypeScript frontend integrated with Spring Boot REST APIs.
- Applied AI-assisted coding for generating implementation options, resolving bugs, improving code structure, and documenting backend/frontend flows.
- Strengthened practical understanding of secure application design, maintainable backend architecture, API integration, and enterprise-style full-stack development.
- Tech Stack: React.js, TypeScript, Spring Boot, Spring Security, JWT, MySQL, Hibernate JPA

InkVision – OCR and Handwritten Character Recognition

- Designed and trained a Convolutional Neural Network using TensorFlow for handwritten digit and alphabet recognition.
- Achieved 97% accuracy on MNIST and A–Z Handwritten Alphabet datasets.
- Built a real-time image preprocessing and prediction pipeline using OpenCV and TensorFlow.
- Developed a Tkinter-based desktop application capable of running fully on local hardware.
- Applied AI-assisted learning and debugging to understand model behavior, improve preprocessing flow, and validate prediction results.
- Gained practical exposure to computer vision, model training, local AI execution, and lightweight AI application deployment.
- Tech Stack: Python, TensorFlow, OpenCV, Tkinter

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, SQL, C

Frontend: React.js, Angular, HTML, CSS

Backend: Spring Boot, Spring Security, FastAPI, Hibernate JPA, REST APIs

Databases and Vector Stores: MySQL, MongoDB, Qdrant

AI / ML: RAG, Agentic AI Concepts, Vector Search, Semantic Search, CrewAI, Ollama, FastChat, TensorFlow, OpenCV

AI-Assisted Development: Codex-style coding agents, Claude-style development assistance, prompt-based coding, debugging, refactoring, documentation support, test ideation

Tools: Git, GitHub, IntelliJ IDEA, VS Code, PyCharm

Operating Systems: Linux, Windows

CERTIFICATIONS

Introduction to Modern AI – Cisco Networking Academy

Python for Data Science – NPTEL

The Joy of Computing Using Python – NPTEL | Score: 92%

Principles of Software Design – Duke University

Demystifying Networking – NPTEL

MongoDB Basics for Students – Badge

CORE ENGINEERING FOCUS

AI-Native Software Development: Using AI coding assistants for faster implementation, code explanation, debugging, refactoring, documentation, and productivity improvement. Agentic AI and Orchestration: Learning and applying agent-based workflows using CrewAI, local LLMs, task decomposition, and tool-based execution patterns. RAG and Personalized Context: Building document ingestion, semantic search, vector retrieval, and personalized context retrieval using Qdrant and embedding-based search. Secure Full-Stack Engineering: Developing backend and frontend applications with Spring Boot, React, JWT, RBAC, REST APIs, and database-driven workflows. Operational Excellence with AI: Exploring how AI can improve developer productivity, IT efficiency, knowledge retrieval, reporting automation, and software maintainability.

ACHIEVEMENTS

Finalist – HackEarth Competition, demonstrating innovation and applied problem-solving capability. Secured 2nd Place – Capture the Treasure security event, Agni College of Technology, showing interest in cybersecurity and analytical thinking. Solved 300+ Data Structures and Algorithms problems on LeetCode, strengthening coding discipline and algorithmic problem solving. Earned 671 medals and 21 certificates on Skillrack platform, reflecting consistency in programming practice. Attended AI Developers' Toolkit – Aptitude Guru Hem, supporting continued learning in modern AI development. Participated in Google Campus Challenge Solution, gaining exposure to solution-oriented technology challenges.

CAREER INTERESTS

Seeking internship or entry-level opportunities in software development, backend engineering, full-stack development, or AI application development, with a focus on building secure, intelligent, and productivity-enhancing software systems using modern AI-assisted engineering practices.